

More about Frobenius

Recall

k : finite field with characteristic p

$$\begin{array}{ccc} \text{Fr} : k & \longrightarrow & k \\ \alpha & \longmapsto & \alpha^p \end{array}$$

is an automorphism of fields.

We call it the Frobenius automorphism.

Key point:

$$(\alpha + \beta)^p = \alpha^p + \beta^p$$

$$\text{Fr}(\alpha + \beta) = \text{Fr}(\alpha) + \text{Fr}(\beta)$$

Uses characteristic p condition

Proposition:

k : finite field, $|k| = p^d$

Then:

$$(a) \quad \text{Fr}^d = \text{id}_k$$

In fact, d is the smallest positive integer for which this is true

(b) For any $m \geq 1$

$$\{ \alpha \in k : \text{Fr}^m(\alpha) = \alpha \}$$

is a subfield of k

$$k_m = k^{\text{Fr}^m}$$

$$(c) \quad |k_m| = p^{\gcd(d, m)}$$

Pf: (a) $\text{Fr}^d(\alpha) = \alpha^{p^d} = \alpha \checkmark$

Conversely, if

$$\text{Fr}^m(\alpha) = \alpha, \forall \alpha \in k$$

then $\alpha^{p^m} = \alpha, \forall \alpha \in k$

$(\Rightarrow)$ $\alpha^{p^m-1} = 1, \forall \alpha \in k^\times$

The fact that $k^\times$ has a generator of order p^d-1

tells us that m must be at least d .

(b) $k_m = \{ \alpha : \text{Fr}^m(\alpha) = \alpha \}$

• (Addition, 0) $\alpha, \beta \in k_m$

• $\text{Fr}^m(\alpha + \beta) = \text{Fr}^m(\alpha) + \text{Fr}^m(\beta)$

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$$\Rightarrow \alpha + \beta \in k_m$$

$$\cdot \text{Fr}^m(0) = 0.$$

• (Multiplication, 1) ✓

• (Inverses) $\alpha \in k_m \setminus \{0\}$

$$\boxed{\text{Fr}^m(\alpha^{-1}) = \text{Fr}^m(\alpha)^{-1}} \\ = \alpha^{-1} \quad \checkmark$$

$$\Rightarrow \alpha^{-1} \in k_m.$$

$$(c) \quad |k_m| \stackrel{??}{=} p^{\text{ord}(m,d)}$$

Pf: We know

$$|k_m| = p^r, \quad \underline{r \mid d}.$$

$\Rightarrow$ Apply part (a) to k_m

$$\begin{aligned}
 \underbrace{k_m}_{\text{has } p^r \text{ elements}} &\subseteq k_r = \{\alpha \in k : \underbrace{\text{Fr}^r(\alpha) = \alpha}\} \\
 &= \{\alpha \in k : \underbrace{\alpha^{p^r} = \alpha}\}_{\text{has at most } p^r \text{ elements}}
 \end{aligned}$$

$$\Rightarrow k_m = k_r$$

To finish, it is enough to know

$$k_m = k_{\gcd(m, d)}$$

$$\begin{aligned}
 \gcd(m, d) &= sm + td \\
 s, t &\in \mathbb{Z}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \text{Fr}^{\gcd(m, d)} &= \text{Fr}^{sm + td} \\
 &= (\text{Fr}^m)^s \circ (\text{Fr}^d)^t \\
 &= (\text{Fr}^m)^s
 \end{aligned}$$

$$\Gamma_{m=2, d=3}, \quad 1 = (-1) \cdot 2 + 1 \cdot 3$$

$$Fr^1 = Fr^{(-1) \cdot 2 + 1 \cdot 3}$$

$$= Fr^{-2} \circ Fr^3$$

$$\underbrace{\quad}_{\parallel} \\ (Fr^{-1})^2$$

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$$\Rightarrow \text{If } Fr^m(\alpha) = \alpha, \text{ then}$$

$$Fr^{\gcd(m,d)}(\alpha) = (Fr^m)^s(\alpha) \\ = \alpha$$

$\Rightarrow$

$$\boxed{K_m \subseteq K_{\gcd(m,d)} \\ \subseteq K_m}$$

$$\left[\begin{array}{l} Fr^{\gcd(m,d)}(\alpha) = \alpha \\ \Rightarrow Fr^m(\alpha) = \alpha \end{array} \right]$$

$$\Rightarrow K_m = K_{\gcd(m,d)}$$

has exactly $p^{\gcd(m,d)}$ elements.

Example:

$$K = \frac{\mathbb{F}_3[x]}{(x^2 - x - 1)} \quad : \quad \text{field of order 9}$$

$$\alpha = x + (x^2 - x - 1) \quad \boxed{\alpha^2 = \alpha + 1} \quad (*)$$

$$K = \{0, 1, -1, \alpha, \alpha+1, \alpha-1, -\alpha, -\alpha+1, -\alpha-1\}$$

$$\text{Fr}(\alpha) = \alpha^3 = \alpha^2 \cdot \alpha$$

$$= (\alpha + 1) \cdot \alpha$$

$$= \alpha^2 + \alpha$$

$$= \alpha + 1 + \alpha = 2\alpha + 1$$

$$= \underline{-\alpha + 1}$$

$$\begin{aligned} \text{Fr}(-\alpha+1) &= -\text{Fr}(\alpha) + \text{Fr}(1) \\ &= (\alpha-1) + 1 \\ &= \alpha \end{aligned}$$

$$\begin{aligned} \text{Fr}(-\alpha) &= \alpha-1 & \text{Fr}(\alpha-1) \\ \text{Fr}(\alpha-1) &= -\alpha &= -[\text{Fr}(\alpha) + 1] \\ & &= -[\alpha+1+1] \\ & &= -[-\alpha-1] \\ & &= \alpha+1 \end{aligned}$$

$$\text{Fr}(\alpha+1) = -(\alpha+1)$$

$$\left\{ 0 \overset{\text{Fr}}{\mapsto} 1, 1 \overset{\text{Fr}}{\mapsto} -1, -1 \overset{\text{Fr}}{\mapsto} \alpha, \alpha \overset{\text{Fr}}{\mapsto} -\alpha+1, -\alpha \overset{\text{Fr}}{\mapsto} \alpha-1, \alpha+1 \overset{\text{Fr}}{\mapsto} -(\alpha+1) \right\}$$

$$\{\alpha, -\alpha+1\} : \text{zeros for } x^2 - x - 1$$

$$\{-\alpha, \alpha-1\} : \text{zeros for } x^2 + x - 1$$

$$\{\alpha+1, -(\alpha+1)\} : \text{zeros for } x^2 + 1$$

MATH 3312, SPRING 2026: LECTURE 10, SIZE OF k_r

The argument given for part (c) of the Proposition was incomplete. To finish, we still have to show:

Fact 1. Suppose that $|k| = p^d$ and $r \mid d$. Then

$$k_r = \{\alpha \in k : \text{Fr}^r(\alpha) = \alpha\}$$

has exactly p^r elements.

Proof. k_r consists of solutions to the equation $\alpha^{p^r} - \alpha = 0$ in k , and so $k_r^\times$ consists of solutions to the equation $\alpha^{p^r-1} - 1 = 0$ in $k^\times$. It is enough to know $|k_r^\times| = p^r - 1$. In the usual way, because this equation involves a degree $p^r - 1$ polynomial, we know that we can have *at most* $p^r - 1$ solutions.

Since $r \mid d$, we know

$$p^r - 1 \mid p^d - 1.$$

See the argument from problem 5 (a) $\Rightarrow$ (b) of Homework 3.

Moreover, $k^\times$ is cyclic of order $p^d - 1$. Therefore, $k^\times$ has an element of order $p^r - 1$: If β is a generator, then $\beta^{(p^d-1)/(p^r-1)}$ will have order $p^r - 1$. Therefore, $k^\times$ contains a cyclic subgroup of order $p^r - 1$, and all the elements α in this subgroup will satisfy $\alpha^{p^r-1} = 1$. In other words, we have *at least* $p^r - 1$ solutions, and now we're done. $\square$